

## **Motion Detector with Buzzer**

### **Hardware Components Required:**

- Raspberry Pi 4 with GPIO extension
- Breadboard
- One Passive Infrared (PIR) motion sensor
- One Active buzzer
- Wires: red, black, yellow, blue colors

### **Step-by-step Instructions to Connect Hardware Components:**

1. Place the PIR motion sensor near the breadboard.
2. Connect the red wire to GPIO 5V (pin 2).
3. Connect the black wire to GPIO GND (pin 6).
4. Connect the yellow wire to GPIO 17 (pin 11).
5. Plug the active buzzer on the breadboard so that each leg is in a new empty row.
6. Connect the buzzer's longer leg (+) to GPIO 27 (pin 13).
7. Connect the buzzer's shorter leg to the GPIO GND pin (pin 6).
8. Power on Pi.
9. On the screen, **click on the projects folder**.
10. Click on the **motion\_sensor.py** file.
11. Write the code.



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### Python code (Solution):

```
1  # motion_sensor.py
2
3  from gpiozero import MotionSensor, Buzzer
4  from time import sleep
5
6  # Define GPIO pins
7  PIR_PIN = 17
8  BUZZER_PIN = 27
9
10 # Initialize devices
11 pir = MotionSensor(PIR_PIN)
12 buzzer = Buzzer(BUZZER_PIN)
13
14 print("PIR Motion Detector active. Stabilizing sensor...")
15 sleep(30) # Give PIR time to calibrate
16 print("System ready. Move in front of the sensor to test.")
17
18 try:
19     while True:
20         pir.wait_for_motion()
21         print("Motion detected! Beeping 3 times...")
22
23         # Beep 3 times when motion is detected
24         for _ in range(3):
25             buzzer.on()
26             sleep(0.2) # short beep
27             buzzer.off()
28             sleep(0.2) # short pause between beeps
29
30         print("Beeping complete. Waiting for no motion...")
31
32         pir.wait_for_no_motion()
33         print("No motion detected. Buzzer OFF.")
34         buzzer.off() # ensure buzzer is off
35
36 except KeyboardInterrupt:
37     buzzer.off()
38     print("\nProgram terminated.")
39
```